

BEIRUT RAFIC HARIRI INTL, Lebanon

WMO: 401000

Lat: 33.821N

Lon: 35.488E

Elev: 27

StdP: 101.01

Time Zone: 2.00 (LBN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.4	9.8	-2.9	3.0	13.8	-0.8	3.5	14.1	14.2	14.4	12.4	14.5	3.5	110	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	32.8	23.4	31.5	24.3	30.9	24.5	26.9	30.6	26.2	30.0	25.8	29.6	3.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.2	30.0	25.1	20.3	29.5	24.5	19.6	29.1	84.9	30.6	82.1	30.2	80.0	29.7	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.4	7.1	6.4	36.7	1.7	1.7	5.2	37.9	4.2	38.9	3.3	39.9	2.1	41.2
			3.1	28.2	1.8	0.8	1.8	28.8	0.7	29.3	-0.3	29.7	-1.6	30.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.5	14.6	14.9	16.7	19.3	22.5	25.4	27.6	28.4	27.2	24.5	20.3	16.4
	DBStd	5.35	2.18	2.40	2.78	2.93	2.38	1.49	1.03	0.95	1.25	2.21	2.50	2.22
	HDD10.0	2	1	1	0	0	0	0	0	0	0	0	0	0
	HDD18.3	381	118	99	67	18	1	0	0	0	0	1	10	67
	CDD10.0	4199	143	138	208	278	387	461	546	569	517	449	308	197
	CDD18.3	1536	1	3	16	47	129	211	288	311	267	191	68	6
	CDH23.3	13694	3	9	89	250	609	1617	3186	3707	2728	1285	203	9
(14) Wind	WSAvg	3.0	3.2	3.2	3.2	3.1	3.0	3.2	3.4	2.9	2.8	2.6	2.6	3.0
	PrecAvg	784	180	128	106	45	16	1	0	0	4	52	110	159
	PrecMax	1157	641	322	263	230	80	11	4	9	30	172	315	329
	PrecMin	352	54	34	10	0	0	0	0	0	0	0	0	25
	PrecStd	215	100	70	58	42	17	3	1	2	7	41	71	85
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	24.8	30.0	32.9	34.2	32.9	33.2	33.7	33.0	33.2	29.2	24.3
		MCWB	14.2	14.4	16.0	17.0	19.0	20.7	25.8	26.7	24.8	20.0	18.2	15.5
	2%	DB	20.2	21.4	25.5	29.0	30.8	30.4	31.6	32.1	31.2	30.8	27.0	22.2
		MCWB	14.1	14.2	15.1	16.9	19.2	23.2	25.8	26.3	24.3	20.9	18.2	15.3
	5%	DB	19.1	19.8	22.9	26.0	28.0	29.1	30.8	31.2	30.3	29.0	25.3	21.0
		MCWB	13.7	14.1	14.9	17.0	19.7	23.4	25.4	25.7	24.0	21.5	18.0	15.0
	10%	DB	18.1	18.5	20.8	23.8	26.1	28.2	30.1	30.8	29.9	27.9	24.1	20.0
		MCWB	13.1	13.5	15.2	17.1	20.3	23.2	25.1	25.4	23.8	21.5	17.6	14.5
	0.4%	WB	16.6	17.2	18.6	20.4	23.4	26.0	27.6	28.0	27.0	25.1	21.6	18.0
		MCDB	19.7	20.6	22.7	25.4	26.8	29.7	31.1	31.6	31.1	28.4	25.7	21.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	15.3	15.9	17.5	19.4	22.3	25.2	26.7	27.1	25.9	23.9	20.4	16.7
		MCDB	18.8	19.3	21.3	24.0	25.7	28.4	30.2	30.9	29.9	27.8	24.8	20.7
	5%	WB	14.4	15.0	16.6	18.6	21.7	24.5	26.2	26.4	25.2	23.0	19.4	15.7
		MCDB	18.1	18.6	20.5	23.0	25.1	27.8	29.7	30.3	29.4	27.2	24.0	19.7
	10%	WB	13.6	14.1	15.7	17.9	21.0	23.9	25.7	26.0	24.6	22.2	18.3	15.0
		MCDB	17.4	17.8	19.8	22.1	24.6	27.4	29.3	29.9	29.0	26.8	23.1	18.9
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.5	6.9	7.4	7.1	6.3	5.8	5.6	5.6	6.1	6.5	6.2
		MCDBR	7.2	8.4	10.7	11.6	10.1	7.5	6.7	6.3	6.4	7.4	7.8	7.4
		MCWBR	5.5	6.2	6.2	6.1	6.0	5.8	5.6	5.3	5.6	6.4	6.6	5.8
	5% WB	MCDBR	6.5	7.4	8.5	8.8	7.6	6.8	6.3	6.2	6.0	6.2	6.9	6.6
		MCWBR	5.5	6.1	6.0	5.7	5.8	5.9	5.6	5.5	5.8	6.1	6.4	5.5
(40) Clear-Sky Solar Irradiance	taub		0.400	0.440	0.493	0.549	0.541	0.462	0.479	0.488	0.467	0.482	0.424	0.406
	taud		2.209	2.080	1.942	1.825	1.885	2.134	2.098	2.070	2.117	2.052	2.180	2.200
	Ebn at Noon		786	791	780	756	768	829	812	798	795	739	751	751
	Edh at Noon		114	144	178	210	200	155	160	162	148	146	117	110
(44) All-Sky Solar Radiation	RadAvg		2.54	3.37	4.83	6.11	7.36	8.15	7.97	7.22	6.08	4.46	3.19	2.40
	RadStd		0.24	0.36	0.32	0.41	0.34	0.14	0.11	0.14	0.16	0.23	0.20	0.24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.43	N/A	-1.45	+0.51	N/A	N/A	N/A	-69	+227	+156
Regional (13 neighbors)	+0.30	N/A	-1.33	N/A	N/A	N/A	N/A	N/A	+173	+110

Nomenclature: See separate page